

# SOP-P201-MAINT-001 Rev.2 · P-201 Pump Maintenance Procedure (2019)

BPCL Mathura Refinery · Rotating Equipment Division

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## Document Control

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Approved by: V.P. Mishra (Head, Maintenance). Next Review Due: March 2022. Status:  
SUPERSEDED · DO NOT USE.

## 1. Scope

This procedure covers planned maintenance activities for P-201 Crude Oil Transfer Pump including bearing replacement, coupling inspection, and mechanical seal servicing at BPCL Mathura Refinery Unit 2.

## 2. Torque Specifications

WARNING: All bolts must be torqued to specification using calibrated torque wrench.

Coupling bolts (M24 grade 8.8): TORQUE TO 80 Nm (as per Flowserve OEM manual 2015 edition). Bearing housing end cover bolts (M16): 45 Nm. Baseplate hold-down bolts (M30): 180 Nm. Casing flange bolts (M20): 95 Nm.

## 3. LOTO Sequence (2019 Version)

LOTO STEP A: First close SUCTION valve V-201A completely (handwheel, 5 turns CW).

LOTO STEP B: Then close DISCHARGE valve V-201B completely (handwheel, 8 turns CW).

LOTO STEP C: De-energize MCC-2 breaker CB-201 and lock with personal padlock. NOTE:

Sequence A then B is critical · do not reverse. LOTO STEP D: Attach danger tags at

V-201A handwheel, V-201B handwheel, and MCC-2. LOTO STEP E: Open vent valve V-201C to release casing pressure.

## 4. PPE Requirements

Chemical resistant gloves (nitrile, minimum 6mil thickness). Safety glasses or face shield. Steel-toed safety boots. Flame-resistant coveralls (Category 2). Hard hat. Hearing protection when pump is running.

## 5. Bearing Replacement

Use bearing puller to remove old bearings. Do not use heat gun unless bearings are seized (heat to maximum 100 degC using induction heater only · no open flame). Clean bearing housing with lint-free cloth and isopropyl alcohol. Press new bearing using hydraulic press or freeze fit (liquid nitrogen, -40 degC). Apply Mobil SHC 100 grease (15g per housing) after installation.